Full support and positive density in the zero-temperature SK model

Two uses of a coercive estimate · A reconstruction of OpenAI's argument · 24 October 2026

Abstract. An integrable minimiser of the zero-temperature Parisi functional for the pure, zero-field Sherrington–Kirkpatrick model has full relative Stieltjes support on $[0,1)$ and a smooth density that is strictly positive on $(0,1)$. We reconstruct OpenAI's argument around the estimate responsible for both conclusions. First variation gives contact at the endpoints of a proposed gap. On a positive constant interval, coercivity makes the derivative of a diffusion moment strictly convex, producing an impossible area deficit; zero coefficients and endpoints have separate arguments. After full support has fixed that moment at every time, the same estimate passes through a fixed-time approximation and forces the density to be positive. We explain why this passage concerns spatial moments rather than time derivatives. Chen had already proved full support and smooth absolute continuity. The contribution here is an account of the two uses of the quantitative estimate, with their analytic inputs and attribution retained.

1. The result and the mechanism

For independent standard Gaussian couplings J_{ij}, the Sherrington–Kirkpatrick optimization problem is to maximise

$$H_n(\sigma) = n^{-1/2} \sum_{i<j} J_{ij}\sigma_i\sigma_j, \qquad \sigma \in \{-1,1\}^n.$$

Write $P_* := \lim_{n\to\infty} n^{-1}\mathbb{E}\max_\sigma H_n(\sigma)$ for the limiting optimum per spin. The Parisi formula expresses this value as a minimum over nondecreasing functions (see Auffinger and Chen (2017), Theorem 1 and proof of Lemma 3). The parameter t ranges over nonnegative overlap scales: the overlap of two configurations is $R = n^{-1}\sum_i \sigma_i\tau_i$. We explain why a minimising function has no interval of constancy, and why its density is positive in the interior.

This is the analytic part of one proof that the quantum approximate optimization algorithm (QAOA) approaches P_* with the system-size limit taken first. Full support supplies the moment identities used to construct a near-optimal classical input for the circuit; positive density is a further conclusion. Section 5 explains this connection and points to the separate circuit argument. The proofs here use Itô calculus and standard martingale convergence, with the required Parisi PDE estimates stated where they enter.

We use pure SK covariance $\xi(t) = t^2/2$ and zero external field. The covariance of H_n is $nR^2/2 - 1/2$; adding one common independent centred Gaussian of variance $1/2$ restores $n\xi(R)$ and leaves the expected maximum unchanged.

Let $\mathcal{U}$ consist of the nonnegative, nondecreasing, right-continuous functions $\gamma : [0,1) \to [0,\infty)$ satisfying $\int_0^1 \gamma(t)\,dt < \infty$. Define the locally finite Stieltjes measure by

$$\mu_\gamma([0,t]) = \gamma(t), \qquad 0 \le t < 1.$$

Thus an atom $\gamma(0)$ is allowed at zero, and the total mass need not be finite or equal to one. All support statements are relative to $[0,1)$.

For this coefficient, let $\Phi = \Phi_\gamma$ be the Parisi value function, the stochastic-control solution of

$$\Phi_t = -\tfrac{1}{2}(\Phi_{xx} + \gamma(t)\Phi_x^2), \qquad \Phi(1,x) = |x|, \qquad \mathcal{P}(\gamma) = \Phi_\gamma(0,0) - \tfrac{1}{2}\int_0^1 t\gamma(t)\,dt.$$

Equivalently, this solution is obtained by the stable approximation of γ by bounded monotone coefficients. The exact control definition and stability theorem are OpenAI (2026a), §1 and Lemmas 2.1–2.2. Write $u = \Phi_x$ and $a = \Phi_{xx}$, and let

$$dX_t = \gamma(t)u(t, X_t)\,dt + dB_t, \qquad X_0 = 0, \qquad q(t) = \mathbb{E}[u(t, X_t)^2],$$

where B is standard Brownian motion. The diffusion exists and is unique for the fixed Brownian motion. We use stochastic calculus on compact time intervals below one; the required analytic inputs are stated in §2.

Chen (2026), Theorem 1.3 had already proved full support and smooth absolute continuity. The following formulation and quantitative proof are from OpenAI; the positive-density consequence is treated separately in §4.

Theorem 1 (full support; OpenAI (2026a), Theorem 1.1). If $\gamma \in \mathcal{U}$ minimises $\mathcal{P}$, then

$$\operatorname{supp}\mu_\gamma = [0, 1), \qquad \gamma(t_2) > \gamma(t_1) \quad (0 \le t_1 < t_2 < 1).$$

For every associated diffusion and every $0 \le t < 1$,

$$q(t) = t, \qquad \mathbb{E}[a(t, X_t)^2] = 1.$$

Its Stieltjes measure has a smooth density, strictly positive at every interior time; Corollary 3 gives the precise stronger statement. Existence of a minimiser is a separate input from Auffinger and Chen (2017), Theorem 1 and the proof of Lemma 3. Integrability allows γ to grow near one. Zero field gives $u(0, 0) = 0$, and $\xi'' = 1$ is used in the curvature calculation; the theorem here does not extend these hypotheses to nonzero field or a general mixed-spin covariance.

The proof compares two demands on a hypothetical gap. Minimality forces q to agree with time at its endpoints, with endpoint slopes at most one. On a positive constant interval a quantitative estimate makes q' strictly convex, so its integral is smaller than the interval's length. This contradicts the required change in q. After excluding all gaps, the identity $q' = 1$ supplies the equation from which the same estimate will force $\gamma' > 0$.

The support conclusion has a history distinct from this proof. Auffinger, Chen and Zeng (2020), Theorem 1 proved infinite support, which can still have gaps: consider $\sum_{k\ge1} 2^{-k}\delta_{1-2^{-k}}$. We follow OpenAI's quantitative argument and credit its antecedents at the steps where they enter.

2. Minimality gives contact; curvature excludes gaps

For an admissible coefficient, Φ is even, convex and one-Lipschitz in space. Its positive-order spatial derivatives are bounded and continuous on each strip $[0, T] \times \mathbb{R}$, $T < 1$; the PDE holds almost everywhere in time. The compact-strip approximation theorem justifies the following Itô calculations even when γ has jumps (see OpenAI (2026a), Lemmas 2.1–2.2). The control method has antecedents in Jagannath and Tobasco (2016) at positive temperature and Chen, Handschy and Lerman (2018), Proposition 2 and Theorem 5 at zero temperature.

Fix a minimiser. Itô's formula cancels the drift of $u(t, X_t)$:

$$du(t, X_t) = a(t, X_t)\,dB_t, \qquad q(0) = 0, \qquad q'(t) = \mathbb{E}[a(t, X_t)^2].$$

In particular q' is continuous across coefficient jumps. The one-sided variation formula is

$$D_+\mathcal{P}(\gamma)[\eta] = \tfrac{1}{2}\int_0^1 \eta(t)(q(t)-t)\,dt, \tag{1}$$

for every integrable direction η for which $\gamma+\varepsilon\eta \in \mathcal{U}$ for all sufficiently small $\varepsilon \geq 0$ (see OpenAI (2026a), Lemma 6.1). Earlier zero-temperature variation and support identities appear in Chen, Handschy and Lerman (2018), Propositions 3–4.

Adding an atom at s changes the cumulative function by $\eta = \mathbf{1}_{[s,1)}$. Thus

$$G(s) := \int_s^1 (q(t)-t)\,dt \geq 0.$$

Here $G(s)/2$ is the first-order cost of the added atom. Conversely, let ν be the restriction of μ_γ to a compact $K \subset [0,1)$. Removing a fraction of this mass is feasible, so

$$0 \leq D_+\mathcal{P}(\gamma)[-\nu([0,\cdot])] = -\tfrac{1}{2}\int_K G(s)\,d\mu_\gamma(s).$$

The change of integration order is justified by $|q-t| \leq 1$ and $\mu_\gamma(K) < \infty$. Since $G \geq 0$, it vanishes almost everywhere for μ_γ, and continuity makes it vanish on $S = \operatorname{supp}\mu_\gamma$. This removes diffuse mass as well as atoms; no density or probability constraint is assumed.

At a support point $s > 0$, G has a minimum. Hence

$$G'(s) = s - q(s) = 0, \qquad G''(s) = 1 - q'(s) \geq 0, \quad \text{so} \quad q(s) = s, \quad q'(s) \leq 1. \tag{2}$$

If $0 \in S$, then $G(0) = q(0) = 0$ and $G(h) = \frac{1}{2}(1-q'(0))h^2 + o(h^2)$; nonnegativity gives the same slope bound at zero. These are the contact conditions of OpenAI (2026a), Lemma 6.2.

Lemma 2 (constant-interval curvature; OpenAI (2026a), Proposition 5.1). Let $\gamma \in \mathcal{U}$, with its associated diffusion and moment q. If $\gamma = c$ on an open interval $I \subset (0,1)$, then for $t \in I$

$$q'''(t) \geq \frac{c^2}{1000}\mathbb{E}[a(t,X_t)^4] > 0 \quad \text{if } c > 0. \tag{3}$$

If $c = 0$, then $q''(t) = \mathbb{E}[\Phi_{xxx}(t,X_t)^2] > 0$.

Minimality is not a hypothesis of this lemma. Its integrated coercive estimate is explained in §3. We first show what it buys.

Suppose (ℓ, r) is a support gap with endpoints in S and $r < 1$. The coefficient is constant there. If its value is positive, strict convexity of q' and (2) give

$$q'(t) < \frac{r-t}{r-\ell}q'(\ell) + \frac{t-\ell}{r-\ell}q'(r) \leq 1 \qquad (\ell < t < r).$$

Integrating contradicts $q(r) - q(\ell) = r - \ell$. The geometric obstruction is an area deficit under q'. If the constant is zero, strict increase gives $q'(t) < q'(r) \leq 1$ and the same contradiction. An initial gap $[0, r)$ with $0 \notin S$, $r \in S$ and $r < 1$ has coefficient zero and no atom at zero. Use $q(0) = 0$ and contact at r, without assuming contact at zero.

A terminal gap cannot use contact at one, which is outside the support domain. On a constant interval ending at one, OpenAI (2026a), Lemma 6.3 instead proves

$$1 - q(t) \geq C\sqrt{1-t}$$

for some $C > 0$ and sufficiently late t. The terminal smoothing keeps u uniformly away from ± 1 in a window of width $\sqrt{1-t}$ around zero. The diffusion density has a uniform positive lower bound on that window, under the assumed plateau; integrating $1 - u^2$ gives the estimate. Thus $q(t) - t \leq (1-t) - C\sqrt{1-t} < 0$ near one, making $G(s) < 0$ there. This contradicts minimality and also excludes empty support. The zero-coefficient heat argument and late-step variation already appear in Chen, Handschy and Lerman (2018), Theorem 6 and its proof. Terminal-plateau exclusion was already recorded by Auffinger, Chen and Zeng (2020), Remark 7.

These cases exhaust the complement of the relatively closed set S. Consequently $S = [0, 1)$, contact gives $q(t) = t$, and differentiation gives $\mathbb{E}a^2 = 1$, including the right-hand value at zero. Moreover, $\gamma(t_2) - \gamma(t_1) = \mu_\gamma((t_1, t_2]) > 0$. This proves Theorem 1 using the stated analytic inputs.

3. The estimate behind strict curvature

On a positive plateau put $r = c\Phi_x$, $v = r_x$, $w = r_{xx}$ and $z = r_{xxx}$. Along the diffusion, the drift and diffusion pairs for r, v, w are $(0, v)$, $(-v^2, w)$ and $(-3vw, z)$. Starting with $c^2 q' = \mathbb{E}v^2$, Itô's formula gives

$$c^2 q'' = \mathbb{E}[w^2 - 2v^3], \qquad c^2 q''' = \mathbb{E}[z^2 - 12vw^2 + 6v^4]. \tag{4}$$

For example, the drifts of w^2 and v^3 are $z^2 - 6vw^2$ and $3vw^2 - 3v^4$, respectively. The negative mixed term is the obstacle. Convexity, which gives $v \geq 0$, does not control it.

The required estimate is an inequality after integration against the actual diffusion density. Begin with a positive nondecreasing finite-step coefficient and terminal smoothing $M^{-1}\log(2\cosh Mx)$, with $M \geq \|\gamma_{\text{step}}\|_\infty$. On a step of height c, the density factors as $f = e^{c\Phi}g$, where g solves the forward heat equation. For $s = -\partial_x \log g$, the source proves joint shape restrictions: on $x > 0$, r and s are increasing and concave, $w/(2r) + v \leq 0$ is nondecreasing, and s/r is nondecreasing. The ratios extend through zero. These are analytic conclusions of OpenAI (2026a), §3, not consequences of the polynomial in (4).

The factorisation of the density gives $f_x/f = r - s$. For a suitable expression F in the shape quantities, define

$$\mathcal{W}(F) = r\partial_x F + (v + r^2 - rs)F.$$

Then $f\mathcal{W}(F) = \partial_x(rfF)$, so the source's tail estimates give $\mathbb{E}_f \mathcal{W}(F) = 0$. The point is to choose F so that subtracting this zero-mean correction from $z^2 - 12vw^2 + 6v^4$ permits a decomposition into a square and terms nonnegative under the joint shape restrictions, with remainder at least $10^{-3}v^4$. The correction changes the integrand while preserving its expectation. The conclusion is

$$\int_{\mathbb{R}} (z^2 - 12vw^2 + 6v^4) f\, dx \geq \frac{1}{1000} \int_{\mathbb{R}} v^4 f\, dx. \tag{5}$$

The exact identity is OpenAI (2026a), Lemma 4.1; the shape hypotheses, correction and boundary justification are assembled in its Proposition 4.2. The long reasoning record, §4.4, "How the signs make a correction with zero expectation useful", defines every auxiliary variable and reproduces the certificate. Thus (5) is a cited analytic input here, with its mechanism exposed; it is not a pointwise assertion for arbitrary v, w, z.

To obtain Lemma 2 for an integrable coefficient, choose positive monotone step approximations retaining the plateau near a fixed interior time, and send the terminal-smoothing parameter to infinity. Spatial derivatives through order four converge uniformly on a fixed strip below one, and the coupled diffusions converge (see OpenAI (2026a), Lemmas 2.1–2.2 and Proposition 5.1). Only v, w, z remain in (5), so auxiliary density-score convergence is unnecessary. Dividing by c^2 gives (3); strictness follows from $\Phi_{xx} > 0$ on the plateau (see OpenAI (2026a), Lemma 2.3). The zero plateau has the separate heat identity in Lemma 2. Its strictness uses a positive diffusion density and the fact that Φ_{xxx} cannot vanish identically: an even quadratic profile would contradict the one-Lipschitz bound and $|x| + O(1)$ growth.

The finite-step Cole–Hopf representation, transformed heat equations and their jump rules have antecedents in Lopatto (2026), Propositions 3.6, 7.1 and 9.1 and Chen (2026), §§3–4. OpenAI's estimate also uses the joint restriction $r\,\partial_x(s/r) \geq 0$, beyond separate monotonicity and concavity. Chen's Proposition 4.2 proves the crossing implication $q''(t) = 0 \Rightarrow q'''(t) > 0$ on a hypothetical positive gap of the minimiser. In contrast, (3) is quantitative and holds throughout every positive plateau of any admissible coefficient. Chen's Appendix C distinguishes earlier shared estimates from the parallel crossing proofs.

4. A second use: the density cannot vanish inside the interval

Full support does not force pointwise positive density. For instance, $\gamma(t) = \int_0^t (s - \frac{1}{2})^2\,ds$ has full Stieltjes support but $\gamma'(\frac{1}{2}) = 0$. The quantitative remainder in (5) rules this out for a minimiser.

Corollary 3 (smooth positive density; OpenAI (2026a), Corollary 6.4). Under the hypotheses of Theorem 1, $\gamma(0) = 0$ and $\gamma \in C^\infty([0,1))$, with derivatives at zero taken from the right. Its Stieltjes measure is $\mu_\gamma(dt) = \gamma'(t)\,dt$, and

$$2\gamma'(t)\mathbb{E}[a(t, X_t)^3] \geq \frac{\gamma(t)^2}{1000}\mathbb{E}[a(t, X_t)^4] > 0 \qquad (0 < t < 1). \tag{6}$$

Only nonnegativity of the density at zero is asserted.

Write $b = \Phi_{xxx}$ and $d = \Phi_{xxxx}$; all spatial factors in expectations below are evaluated at (t, X_t). Since $da = -\gamma a^2\,dt + b\,dB$ and $\mathbb{E}a^2 = 1$,

$$\gamma(t) = \frac{\mathbb{E}b^2}{2\mathbb{E}a^3}. \tag{7}$$

Initially (7) holds almost everywhere. Its denominator is at least two, because $a \geq 0$ and $\mathbb{E}a^3 \geq (\mathbb{E}a^2)^{3/2} = 1$. The moments are continuous; right continuity of γ extends (7) to every time and proves continuity of γ. Evenness gives $b(0,0) = 0$, hence $\gamma(0) = 0$.

For smoothness, apply Itô's formula to all polynomial moments of finitely many spatial derivatives at once. The derivative of each moment is another such moment plus γ times such a moment. Continuity makes these moments C^1; (7) then makes γ C^1. Repeating proves smoothness without presupposing time derivatives of the coefficient. This is the quotient-and-moment argument of Chen (2026), Proposition 4.6 and Appendix B.6, using the polynomial-moment method of Auffinger and Chen (2015), Lemma 2 and the proof of Theorem 2(ii).

Now the two differentiated moments are

$$\frac{d}{dt}\mathbb{E}b^2 = \mathbb{E}[d^2 - 6\gamma ab^2], \qquad \frac{d}{dt}\mathbb{E}a^3 = \mathbb{E}[3ab^2 - 3\gamma a^4].$$

Differentiating $\mathbb{E}b^2 = 2\gamma\mathbb{E}a^3$ gives

$$2\gamma'\mathbb{E}a^3 = \mathbb{E}[d^2 - 12\gamma ab^2 + 6\gamma^2 a^4]. \tag{8}$$

The expression from the plateau calculation has returned. The minimiser is not constant on an interval, so applying the plateau lemma directly would be invalid. Here is the needed transfer.

Fix $t \in (0,1)$ and $c = \gamma(t) > 0$. For $0 < \delta < \min(t, 1-t)$ define

$$\gamma_\delta(s) = \begin{cases} c, & t-\delta \le s < t+\delta, \\ \gamma(s), & \text{otherwise.} \end{cases}$$

Both joins preserve monotonicity, and this remains an admissible coefficient. Moreover,

$$\|\gamma_\delta - \gamma\|_1 \le 2\delta\,[\gamma(t+\delta) - \gamma(t-\delta)] \longrightarrow 0.$$

Use a subscript δ for its value-function derivatives and diffusion. Lemma 2 and (4), already extended to integrable coefficients, give at the fixed time t

$$\mathbb{E}[d_\delta^2 - 12c\,a_\delta b_\delta^2 + 6c^2 a_\delta^4] \ge \frac{c^2}{1000}\mathbb{E}[a_\delta^4]. \tag{9}$$

Choose a single T with $t < T < 1$. Compact-strip derivative stability and convergence of the coupled diffusions pass each spatial moment in (9) to its counterpart for γ; the estimates do not depend on the plateau width (see OpenAI (2026a), Lemmas 2.1–2.2 and Corollary 6.4). Combining the limiting inequality with (8) proves (6). Strictness follows from $c > 0$ and $\mathbb{E}a^4 \ge (\mathbb{E}a^2)^2 = 1$. In particular, Cauchy–Schwarz gives $\mathbb{E}a^3 \le \sqrt{\mathbb{E}a^4} \le \mathbb{E}a^4$, so the same inequality implies $\gamma'(t) \ge \gamma(t)^2/2000$ for $0 < t < 1$.

We have passed spatial moments, not the time derivatives q_δ'''. Indeed the limiting minimiser has $q(t) = t$ and $q''' = 0$; the coefficient derivative in (8) is exactly what the constant-interval formula omits. The long record, §§5.1–5.2, retains the full moment bootstrap and fixed-time transfer.

5. The scalar consequence and a separate circuit theorem

Full support also recovers the scalar ground-state value P_* defined in §1, without using the density corollary. For a minimiser, the ground-state Parisi formula gives $P_* = \mathcal{P}(\gamma)$ (see Auffinger and Chen (2017), Theorem 1).

The bounded martingale $M_t = u(t, X_t) = \int_0^t a\,dB$ has $\mathbb{E}M_t^2 = t$. It converges almost surely and in L^2 to a variable U with $|U| = 1$. The bound $|\gamma u| \le \gamma$ gives $X_t \to X_1$ almost surely and in L^2. The terminal estimate

$$0 \le \Phi_\gamma(t,x) - |x| \le \sqrt{2(1-t)/\pi} + \tfrac{1}{2}\int_t^1 \gamma(s)\,ds$$

and the convex difference-quotient bounds force $u(t, X_t) \to \operatorname{sign}(X_1)$ on $\{X_1 \neq 0\}$. Thus $UX_1 = |X_1|$, also when $X_1 = 0$. Itô's formula for $\Phi(t, X_t)$ and passage to time one yield $\mathcal{P}(\gamma) = \mathbb{E}|X_1| - \int_0^1 t\gamma(t)\,dt$. Multiplying $X_1 = B_1 + \int_0^1 \gamma M_t\,dt$ by U then cancels the penalty, because $\mathbb{E}[UM_t] = t$:

$$P_* = \mathbb{E}[UB_1] = \int_0^1 \mathbb{E}[a(t, X_t)]\,dt. \tag{10}$$

Fubini uses $|UM_t| \le 1$ and $\gamma \in L^1$; the final cross-isometry uses $\mathbb{E}\int_0^1 a^2\,dt = 1$. This first-moment integral is different from $\mathbb{E}a^2 = 1$. The argument is OpenAI (2026a), Corollary 7.1; the long record, §6, "Recovering the value from a martingale", supplies the endpoint details. The gradient martingale and moment identities have antecedents in Montanari, preprint v2, Lemmas 3.2–3.4, under its positive-temperature support assumption. The zero-temperature martingale and value calculations are in El Alaoui, Montanari and Sellke, preprint v1, Lemmas 6.5 and 6.13. These are method antecedents, distinct from the finite-Gaussian construction below.

Here is the role of full support in the circuit proof. The identity $\mathbb{E}a^2 = 1$ permits normalised, curvature-weighted Gaussian steps, while (10) identifies their target value. At fixed $T < 1$, discretising the diffusion and rounding its terminal gradient with an independent random variable produces a Boolean function of finitely many Gaussians. The rounding preserves the gradient as its conditional mean. A specified sum of products of its correlations with the driving Gaussians and the normalised increments converges to $\int_0^T \mathbb{E}a(t, X_t)\,dt$ as the mesh is refined; taking $T \uparrow 1$ then brings its value to P_*. This is the additional finite construction of OpenAI (2026a), Proposition 7.2, reconstructed in the long record, §7, "A finite Gaussian spin and its adjacent correlations".

There is also a separate circuit theorem. QAOA alternates cost evolution and transverse-field mixing from the uniform superposition (see Farhi, Goldstone and Gutmann (2014)). For every $\varepsilon > 0$, OpenAI (2026b), Theorem 1.1 constructs a finite depth and deterministic angles, independent of system size n and disorder, whose limiting expected energy per spin is at least $P_* - \varepsilon$. The angles and depth may depend on ε; the limit $n \to \infty$ is taken with them fixed. Fixed-parameter limits are due to Farhi, Goldstone, Gutmann and Zhou (2022), whose arbitrary-fixed-angle theorem uses this Hamiltonian and gate convention directly. The eventual-optimality conclusion was conjectured by Basso et al. (2022), §7, equation (7.2). The circuit argument must preserve that correlation value, not just reproduce a scalar number. It realises the finite Gaussian construction on a rooted space, approximates the bounded output by quantum gates, removes auxiliary random seeds, and transfers the resulting single fixed word to SK. These are additional theorems, reconstructed in the long record, §9, "From the scalar formula to one ordinary QAOA word", from OpenAI (2026b), §§2–6 and Appendix A. Neither positive density nor (10) alone proves them. The selected construction gives no quantitative depth bound, finite-size rate or efficient angle-selection procedure.

6. Source and formal scope

This AI-assisted article reconstructs the cited results; it asserts no new theorem. The original analytic proofs are in OpenAI's manuscripts, with the prior mathematical sources identified above. The central imported input is the integrated estimate (5), including its analytic shape and approximation hypotheses.

The selected support implementation, challenge and comparison configuration concern measure and diffusion existence, full support, strict increase and both moment identities, conditional on a minimiser. The scalar implementation, challenge and configuration select a separate formal interface: its scalar and finite-Gaussian clauses assume a minimiser and an associated diffusion, while its ground-state/Parisi equality assumes only the minimiser. Their formal models have not been transported into one another in this record. Neither selected statement includes Corollary 3 or QAOA convergence.

The formal developments use Lean and its mathematical library, mathlib; we credit the system description by de Moura and Ullrich (2021) and The mathlib Community (2020). These scholarly

references describe the tools; the accompanying source record identifies the exact versions used for the selected developments.

Source-integrity checks passed. After an earlier unavailable Comparator startup, later invocations built and exported the admitted full-support and scalar-value challenges. Both stopped during the implementing solution build because imported compiled modules were missing. No successful replay of the corrected build configuration is recorded. Challenge export, tool-fixture success and the exact polynomial calculation establish no research-target acceptance, transitive axiom audit or independent human review. Exact versions, the failed attempts and remaining obligations are retained in the companion source review.

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